

QUICK ACTIVITY 1: SCHOOL SUPPLIES CALCULATOR

A notebook costs ₱45, and a pen costs ₱12. A student buys several notebooks and pens and pays a certain amount. Develop an algorithm that calculates the total cost and change. Assume that quantities are positive whole numbers and the payment is enough to cover the purchase.

DECOMPOSITION

Input	Process	Output
Number of notebooks and pens, and the cash given	Calculate the notebook amount, pen amount, add total amount, change	Present the total amount and change

PSEUDEOCODE

START

 READ notebooks, pens, payment

 notebook_amount $\leftarrow$ notebooks $\times$ 45

 pen_amount $\leftarrow$ pens $\times$ 12

 total_amount $\leftarrow$ notebook_amount + pen_amount

 change $\leftarrow$ payment - total_amount

 DISPLAY total_amount

 DISPLAY change

END

DRY RUN CALCULATIONS

Test Case 1

Inputs:

Notebooks = 3

Pens = 2

Payment = ₱200

Step	Variable	Calculation	Solution
1	Notebook Amount	3×45	₱135
2	Pen Amount	2×12	₱24
3	Total Amount	$135+24$	₱159
4	Change	$200-159$	₱41

Output:

Total amount = ₱159, Change = 41

Test Case 2

Inputs:

Notebooks = 2

Pens = 3

Payment = ₱150

Step	Variable	Calculations	Solution
1	Notebook Amount	2×45	₱90
2	Pen Amount	3×12	₱36
3	Total Amount	$90+36$	₱126
4	Change	$150-126$	₱24

Output:

Total amount = ₱126, Change = ₱24